





Example 1

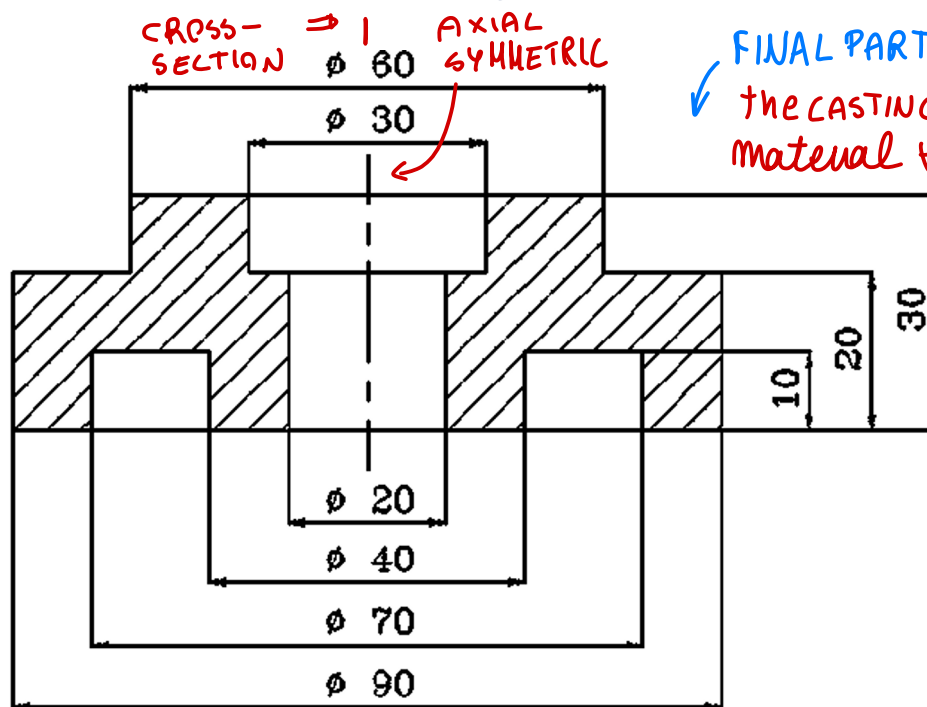


The figure shows a component to be obtained through a sand casting and machining processes. The component will be made of grey cast iron. The pattern will be made of wood.

After having appropriately chosen the parting line, design the pattern and any cores respecting machining allowance (3 mm), draft angles (4°), fillets/radii (3 mm), and shrinkage allowance (1%). Suppose that all surfaces need to undergo subsequent machining.

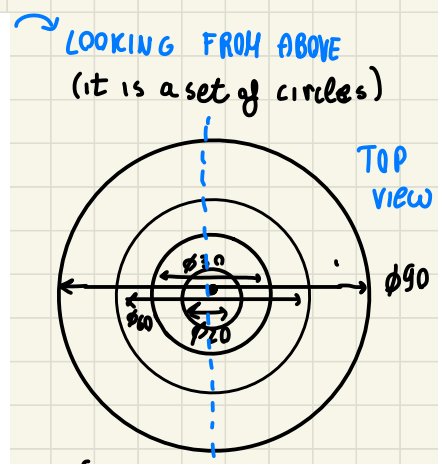
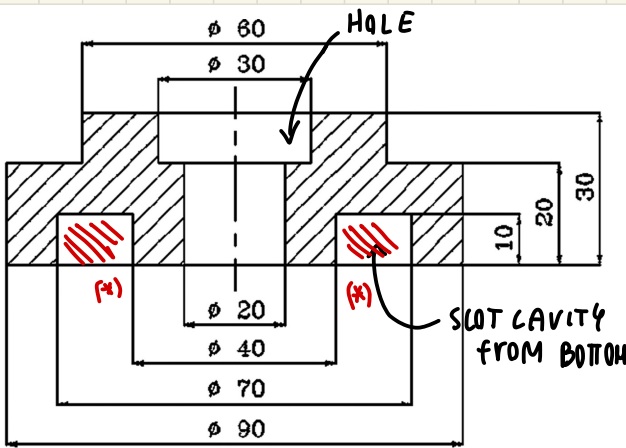
all surfaces need
machining
(allowance $\rightarrow$ 3mm)

$\downarrow$
machining
allowance $\equiv$ material to
add on
surfaces
that
we need
to machine

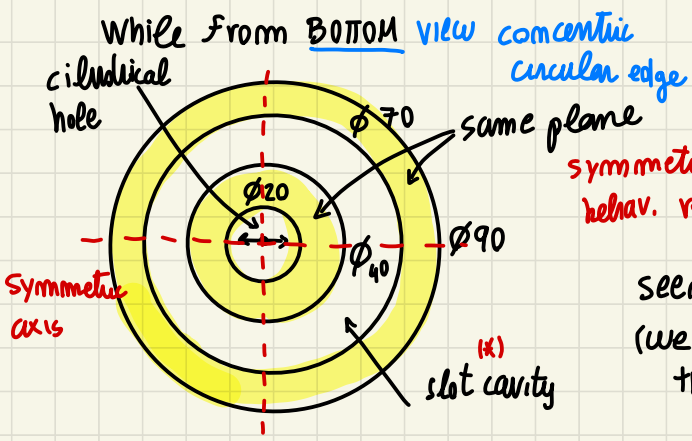


the CASTING may have additional
material & surface

looking @
this design
we have set of
diameters ϕ
and high



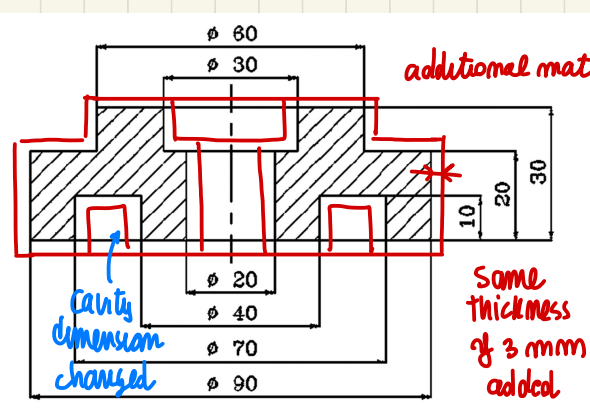
from Above we see that circle (symmetric figure)



seem edges as full line (we can "dash-line" edges that can't be seen)

m.a = 3 mm

MACHINING → on each surface, the final part should have additional material



↙ surface
also on INTERNAL cylinder
external cylinder
↳ adding material = diameter bigger
internal cylinder
reducing diameters
shape of the object after CASTING

We want that path...

We need to create the cavity and internal features (holes)
and create the HOLE

obtainable thanks to the CORE (to design and built)



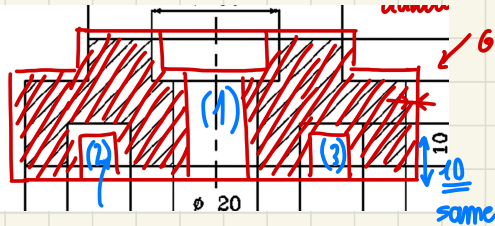
look to final dimension of HOLE to design the CORE
(core not able to resist liquid pressure during pouring / solidification)



with limitations on smallest HOLE / WALLS that you can make
here we suppose possible

CORE { 1: MIDDLE
2: } No CORE, made directly by design DRAG COPE of
3: the mold (depending on where is parting line)

Just the shape of SAND MOLD



DIAMETERS

$$\phi 60 = 60 + 3 \cdot 2 = 66 \text{ mm}$$

external Diam. increase

$$\phi 30 = 30 - 3 \cdot 2 = 24 \text{ mm}$$

internal Diam. decrease

$$\phi 20 = 20 - 3 \cdot 2 = 14 \text{ mm}$$

$$\phi 40 = 40 + 3 \cdot 2 = 46 \text{ mm}$$

$$\phi 70 = 70 - 3 \cdot 2 = 64 \text{ mm}$$

$$\phi 90 = 90 + 3 \cdot 2 = 96 \text{ mm}$$

New dimensions
to satisfy requirement

TO ANSWER THE PROBLEM!

add 3mm on surface
added 2 times on right / left
(the diameter changes)

Machining
allowance

LENGTH

10 = distance between internal / external plane
 $10 + 3 - 3 = 10 \text{ mm}$
+ 3 mm - 3 mm
CAVITY REDUCED
No changes

$$20 = 20 + 3 \cdot 2 = 26 \text{ mm}$$

Added on both sides

$$30 = 30 + 3 \cdot 2 = 36 \text{ mm}$$

FUTURE
machin.

CASTING DIMENSION to have + 3mm V surface

Goal object because of machinery...

considering that object → we need to design the cavity in the mold



SAND CASTING

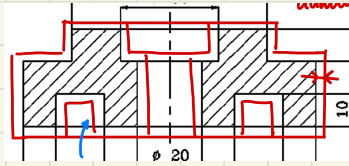
PATTERN, MOLD → to be divided by a PARTING

LINE in $1/2 \cdot 2$ } COPE → upper
DRAIN → below

We must decide

which part is on COPE and DRAG

(different mold part)

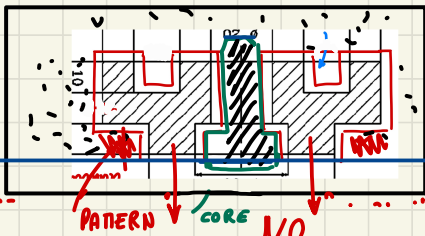


to make this decision, STARTING POINT is the PARTING LINE choice

there can be multiple proposal

PARTING LINE CHOICES (#)

1)



Theoretically
core print space → to position the
CORE inside mold

← with this one we cannot
extract the PATTERN after the
packing of SAND inside mold

CORE designed,

(inside hole will NOT be a pattern
cavity) →

remove that by a CORE PRINT
(full of material)

considering top plane

pattern
must be
removed to
create the
cavity

HOLD flask
fill with
sand around
pattern
(packing sand
around)

→ basic of mold making, (mold must be moved respect
packing, orthogonal movement
respect the plane)

move the drag upside...

DESTROY MATERIAL! → cannot extract the
pattern on that way

to move DRAG respect PATTERN → portion of the sand below pattern,
PORTION that can't be removed

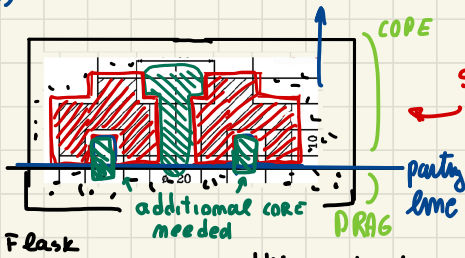
Overhanging
don't allow pattern
removal from MOLD

of MOLD

OVERHANG PART OF
PATTERN respect MOLD

... We need a different PARTING LINE solution without overhangs

2)



all geometry inside the cope

packing sand around

on this situation... ~ you can pack sand above parting line plane

try to move the cope, it

can be moved BUT CIRCULAR SLOT are NOT shaped on that way!

here to CREATE CIRCULAR slot we need to CREATE CORES

CIRCULAR RING-like CORE position on the holes

We need to shape DRAG / CORE

2: FEASIBLE BUT additional CORE needed!

mold bulky

think about DRAG and parting line as upside down

DRAG full of material... you create core print corresponding to the area to not fill... pack material on top

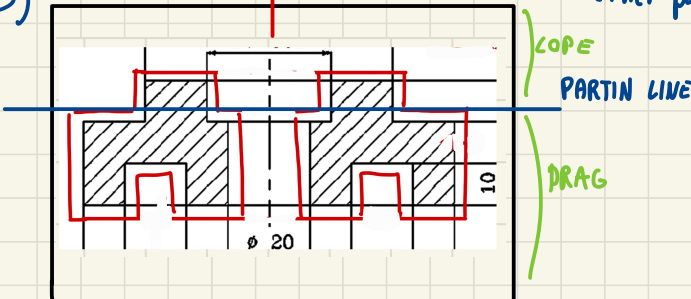


← RING like CORE to shape the holes...

3)

direction of separation

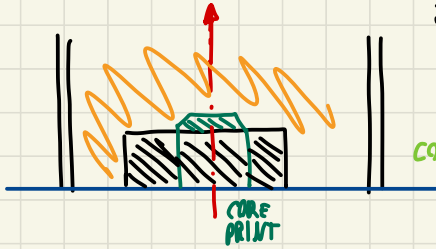
other possible solution...



distributing the shape... between the two mold part

PM OPTION 3 ...

starty from the COPE (upper portion)



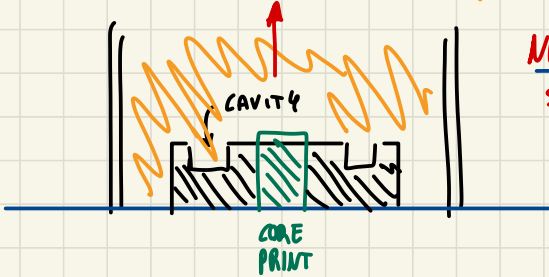
JUST as full
cylinder + CORE PRINT
to future position
THE CORE in the middle

While
the DRAG

pack sand all around
the pattern

NO CORE

↓
Better solution:
less cost!
more convenient

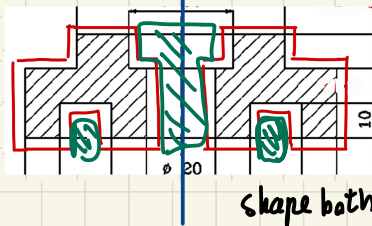


NO PROBLEMS in
separating pattern

↓
SO GOOD PARTING
LINE!

IN THEORY we can change the P.O.V and consider a different
solution

↙ plane across part axis ⇒ make cavity on
different
directions



you'll need more
CORES

shape both COPE/DRAG higher FLASK

Costly CORE!

AXIAL SYMMETRICAL obj

↓
plane through axis symmetry

considering MORE CORES as cavity needed...

← POSSIBLE!
BUT CARE

WE CONSIDER OPTION 3



add an ISSUE... When separating COPE / DRUG respect PATTERN:

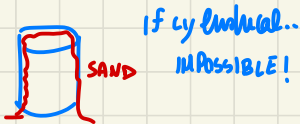
there is an ISSUE



the orthogonal surface with parting plane

same direction of extraction, so material stay in contact with pattern... friction

Increasing risk to damage the SURFACE
(like making beach CASTLES)



IMPOSSIBLE!



so when moving up...
NO FRICTION → easy to separate
VAT from sand

|| Used in sand ||
Casting

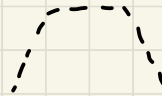
- Vertical plane respect to parting line has a certain slope → **DRAFT ANGLES**

SCOPE created by adding material

We need to have more

material → then it will be

removed by machining



In reality the shape has a certain shape like cylinder → cone

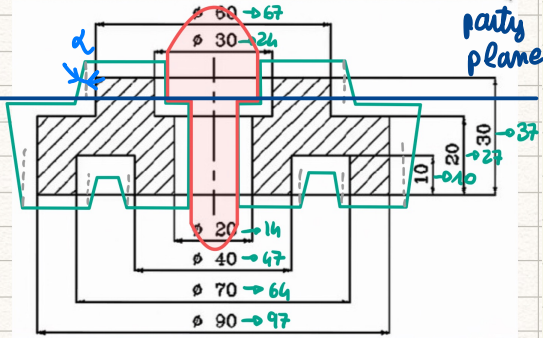
DRAFT ANGLE Needed

↓
easy separation
of pattern respect
cope-drag

DRAFT ANGLE essential on
sand casting

↓
but also another processes where
we have to separate

- PATTERN from sand casting
- permanent mold from final casting



able to REMOVE

α DRAFT ANGLE

to have easier separation

↓
ADDING MATERIAL!

to Design the CORES... We need to consider also

SHRINKAGE ALLOWANCE

machining allowance →
material add to remove after

so multiply dimension
per 1.01

↓
to have desired final
dimension → shrinkage
@ SOLID STATE

$$S.A = 1\%$$

$$\phi 60 = 60 + 3.2 = 66 \text{ mm} \times 1.01 = 66.66$$

$$\phi 30 = 30 - 3.2 = 24 \text{ mm} \times 1.01 = 24.24$$

$$\phi 20 = 20 - 3.2 = 14 \text{ mm} \times 1.01 = 14.14$$

$$\phi 40 = 40 + 3.2 = 46 \text{ mm} \times 1.01 = 46.66$$

$$\phi 70 = 70 - 3.2 = 64 \text{ mm} \times 1.01 = 64.64$$

$$\phi 90 = 90 + 3.2 = 96 \text{ mm} \times 1.01 = 96.96$$

↑

CAREFUL of
HOW I ROUND!

We need TO ROUND, design pattern using [mm]

Wood pattern $\rightarrow$ TO ROUND dimension, be careful on the situation!

Better to: add material that we can Remove later

if NOT enough material $\rightarrow$ cannot remove issue..

{ external dimension ϕ INCREASED
internal dimension ϕ REDUCED

external: CEILING

$$\left\{ \begin{array}{l} \phi 60 = 60 + 3 \cdot 2 = 66 \text{ mm} \times 1.01 = \lceil 66.66 \rceil = 67 \text{ mm} \\ \phi 30 = 30 - 3 \cdot 2 = 24 \text{ mm} \times 1.01 = \lfloor 24.24 \rfloor = 24 \text{ mm} \\ \phi 20 = 20 - 3 \cdot 2 = 14 \text{ mm} \times 1.01 = \lfloor 14.14 \rfloor = 14 \text{ mm} \\ \phi 40 = 40 + 3 \cdot 2 = 46 \text{ mm} \times 1.01 = \lceil 46.46 \rceil = 47 \text{ mm} \\ \phi 70 = 70 - 3 \cdot 2 = 64 \text{ mm} \times 1.01 = \lfloor 64.64 \rfloor = 64 \text{ mm} \\ \phi 90 = 90 + 3 \cdot 2 = 96 \text{ mm} \times 1.01 = \lceil 96.96 \rceil = 97 \text{ mm} \end{array} \right.$$

internal: FLOOR

same for length dimension

$$10 \times 1.01 = 10.10 \text{ , internal dimension} \rightarrow \lfloor 10.10 \rfloor = 10 \text{ mm}$$

$$26 \times 1.01 = \lceil 26.26 \rceil = 27 \text{ mm}$$

$$36 \times 1.01 = \lceil 36.36 \rceil = 37 \text{ mm}$$

} external dimension

Always better to add material instead of removing!

$\rightarrow$ move material

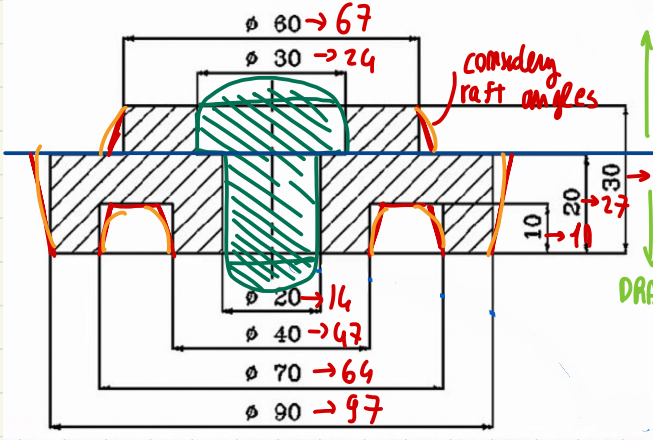
$\downarrow$
removed later
by machining

Not enough material brings to PROBLEM

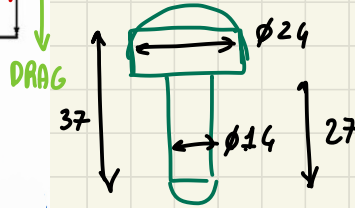
NOW considering the PATTERN with new dimensions ...

new dimensions (before DRAFT ANGLE)
more adding material!

PATTERN final
dimension



CORES
"2 cylinders
+ CORE PRINT"



Sometimes we consider ALSO
RADIUS FILLING $\rightarrow$ "ROUND" the CORNERS

Cause during POURING we have flow of high Temp molten metal
with quite velocity $\rightarrow$

If you have sharp angles edge of SAND
you can DAMAGE IT

$\downarrow$
by rounding you reduce possible damage on
the surface

small

design
variation

on angles + radius

@ end

COPE

DRAW

CORE

to shape

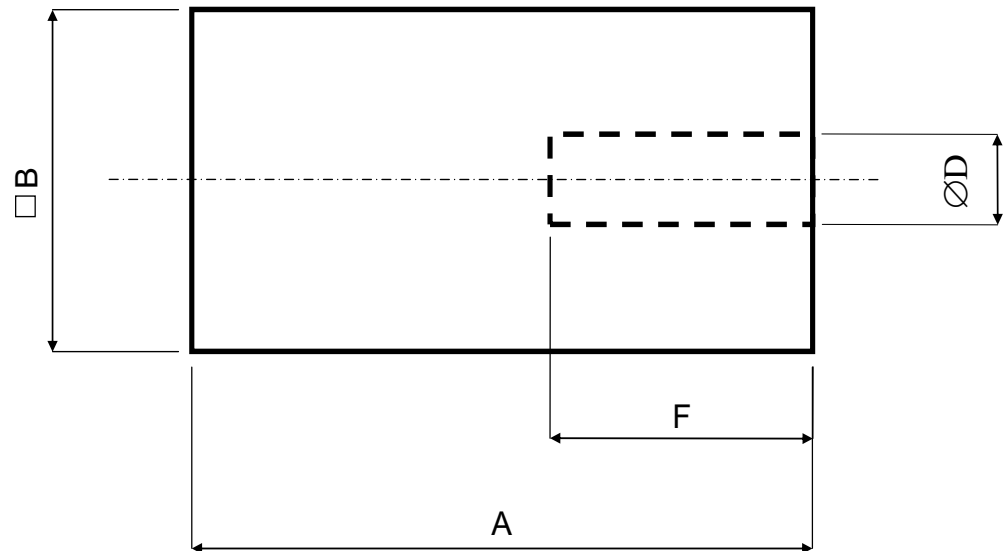
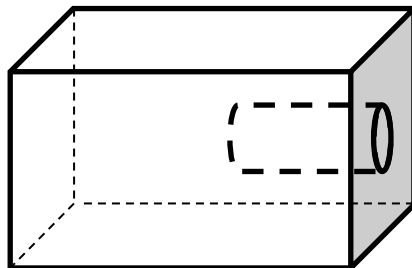


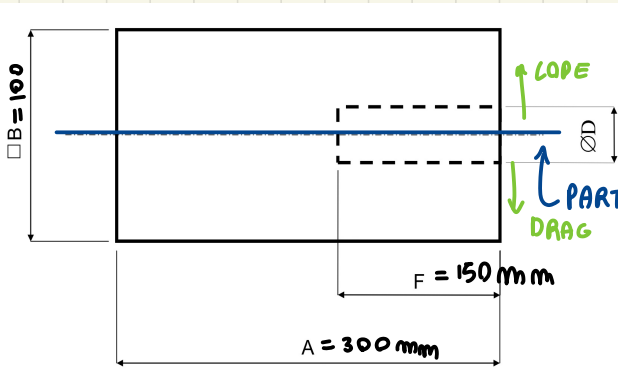
Example 2

The component in the figure must be made by sand casting.

Dimensional data: $A = 300$ mm, $B = 100$ mm, $D = 30$ mm, $F = 150$ mm.

- Knowing that the hole will have to undergo machining, design the pattern assuming a **material allowance** of 2 mm where necessary, and a **shrinkage allowance** of 1%
- If you wish to use a cylindrical core print with a diameter of 70 mm, determine its minimum length to ensure its stability during the forming process





$$m.a = 2 \text{ mm}$$

$$s.a = 1\%$$

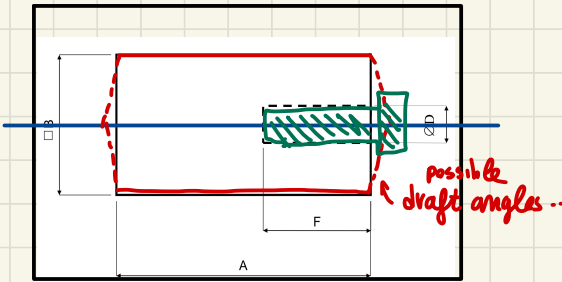
PARTING LINE to design
the MOLD axis of hole
as parting line

(square based block as PATTERN)

dimensions does NOT
need to change RESPECT
machining allowance

↓
NOT removing
material OUTSIDE

↓
JUST the HOLE is
changing dimensions



consider: { shrinkage allowance on full part!
machining on the hole

$$A = 300 \rightarrow 300 \times 1.01 = 303 \text{ mm}$$

$$B = 100 \rightarrow 100 \times 1.01 = 101 \text{ mm}$$

} BLOCK DIMENSION → pattern dimension

We need to PUT a CORE

made of 2 cylinders



space
needed to create
inside mold, to position
THE CORE
↓
core positioned

IF CENTRAL GRAVITY

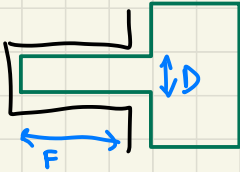
IS OUTSIDE the core print → It will go down → NOT STABLE

(Not able to assemble the mold!)

question (b) is related to CORE
PRINT dimension

DIMENSION the CORE

from $D = 30\text{mm}$ @ beginning



We need m.a $\rightarrow$

$$\text{so } D = 30 - 2 \cdot 2 = 26 \text{ mm}$$

$$D = 26 \times 1.01 = \lfloor 26.26 \rfloor = 26$$

INTERNAL
FEATURE, L J FLOOR

$F = 150$ $\rightarrow$ just add material
on one side

$$F = 150 - 2 + 0 = 148 \text{ mm} \times 1.01 = \lfloor 149.48 \rfloor = 149 \text{ mm} \leftarrow$$

INSIDE $\uparrow$ $\uparrow$ OUTSIDE

this is equal
to CORE
DIMENSION

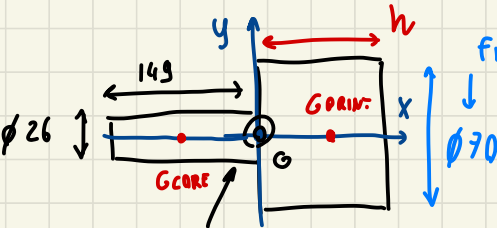
internal geometry

While CORE PRINT (2 cylinders)

b) how to choose
STABLE CORE POSITIONING CORRECTLY h

From the Data

Reference system to position
CENTER OF GRAVITY



We want
here the center of GRAVITY

center of GRAVITY on
 X AXIS

$$G_{\text{PRINT}} = \left(\frac{h}{2}, 0 \right) \quad "CP" \text{ (CORE PRINT)}$$

$$G_{\text{CORE}} = \left(-\frac{149}{2}, 0 \right) \quad "C"$$

$$G = \left(\cancel{X_G}, 0 \right)$$

$\hookrightarrow$ WE WANT $X_G \geq 0$
TOTAL GRAVITY CENTER

$$X_G = \frac{V_C X_C + V_{CP} X_{CP}}{V_C + V_{CP}} \geq 0$$

so that
everything is STABLE!

center of GRAVITY
inside CORE PRINT

to be
sure
 $h \geq 56$

$$V_C X_C + V_{CP} X_{CP} \geq 0 \Rightarrow \frac{\pi}{4} (26)^2 149 \left(-\frac{149}{2} \right) + \frac{\pi}{4} 70^2 h \frac{h}{2} \geq 0$$

$$\Rightarrow h \geq \sqrt{\frac{26^2 149^2}{70^2}} = \lfloor 55.34 \text{ mm} \rfloor = 56 \text{ mm}$$

we ceil that value



Example 3

DIRECTIONAL SOLIDIFICATION study

↳ 2 identical cubes body + RINGS → 3 lanc element separation
⇒ we need to evaluate MODULUS

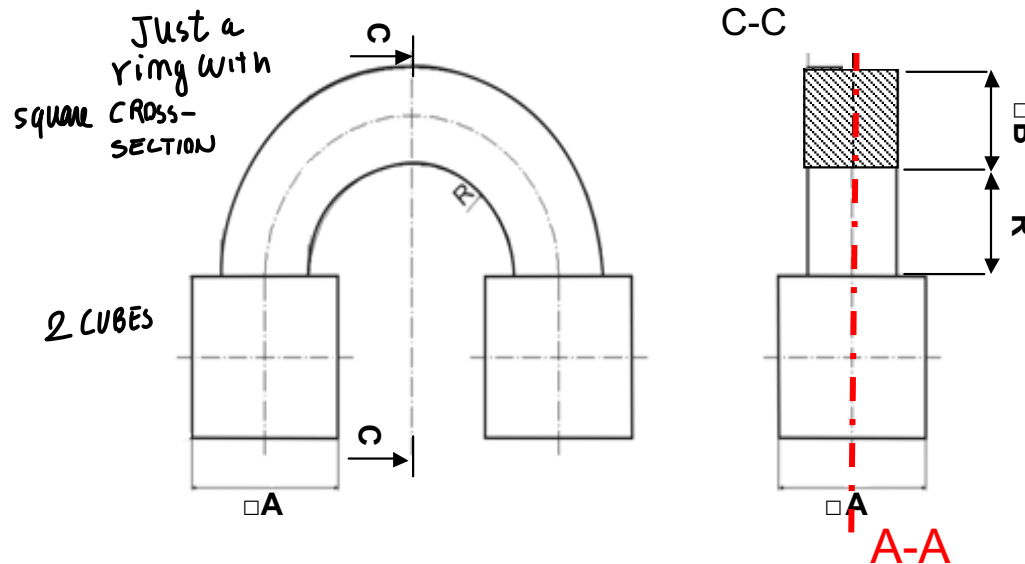
$H = V/A$ { V: volume of element
A: area of shape in contact with the mold material !

The pattern shown will be used to create the cavity in sand casting.

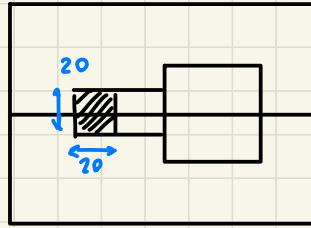
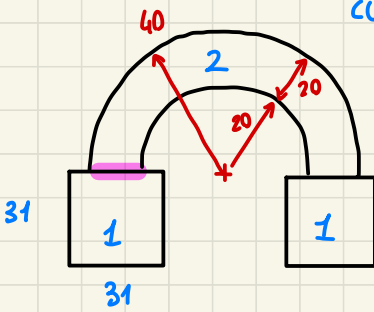
The plane A-A is chosen as the parting line of the mould.

Please, study the **directional solidification** of the metal.

Dimensional data: A= 31 mm, B= 20 mm, R= 20 mm.



cukig ring 20x20, cube 31x31

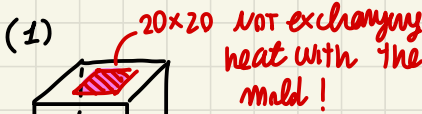


DIRECTIONAL SOLIDIFICATION:

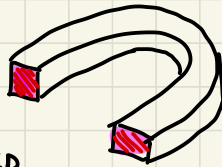
geometric value

by considering the MODULUS: $M = V/A$ of two bodies on contact with mold

on space



(2)



Area parts NOT exchanging heat with the mold

MODULUS evaluation

Body 1.

$$V_1 = 31^3 = 29791 \text{ mm}^3$$

$$A_1 = (31 \times 31) \times 6 - 20^2 = 5366 \text{ mm}^2$$

6 FACES

$$\rightarrow M_1 = \frac{V_1}{A_1} = 5.55 \text{ mm}$$

Body 2. { external + internal cylindrical surface } Area
2 planar rings

$$V_2 = \frac{1}{2} (\pi 40^2 \cdot 20) - \frac{1}{2} (\pi 20^2 \cdot 20) = 37699 \text{ mm}^3 \rightarrow M_2 = \frac{V_2}{A_2} = 5 \text{ mm}$$

$$A_2 = \frac{1}{2} (2\pi 40 \cdot 20 + 2\pi 20 \cdot 20) + \frac{1}{2} (\pi 40^2 - \pi 20^2) = 7540 \text{ mm}^2$$

external ring internal ring circular sections

$$H_1 = 5.55 \text{ mm}$$

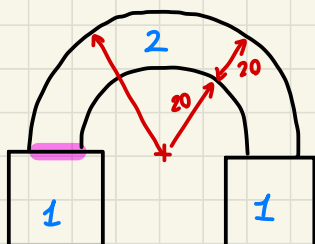
$$H_2 = 5$$

$$\rightarrow \underline{H_2 < H_1}$$

Body 2 solidify first

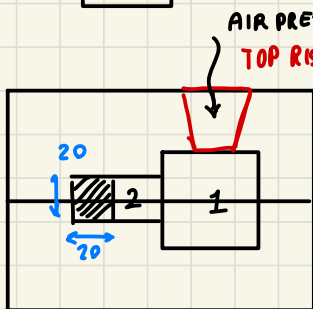
Solidification STARTS on B2 $\rightarrow$ and goes to B1

(1) solidify after (2)

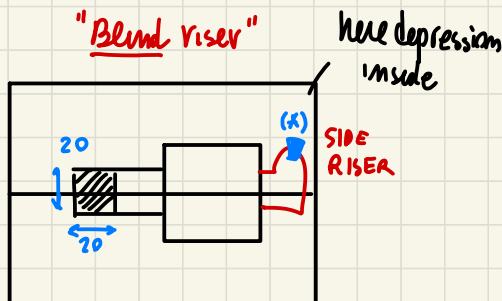


to design the RISERS we need
to have risers
on the top of the
Bodies 1
(TOP/SIDE) riser

air
pressure
to let
material
on R
fill
the cavity



OR



"open riser"
on top

Inside the mold
Not open to OUTSIDE

$\downarrow$
We need to add
RISERS on the top of
BOTH the bodies 1

$\downarrow$
solidification starts on (2)
then from (1) flow on (2)'s cavity
and from (R) to body (1)

or we
will
have
a VACUUM! (x)

When material move
from R $\rightarrow$ depression, so
we need additional
material that create
Gas maintain with
molten metal $\rightarrow$ to have
pressure

to be sure about good behav. (flow correctly)

$$M_1/M_2 = 5.55/5 = 1.11 \in [1, 1 \div 1.3]$$

INSIDE Good range $\rightarrow$

for guarantee
B2 filled
by B1 correctly

RISERS
designed as $M_R > M_1$

usually $M_R \approx 1.2 M_1$ design

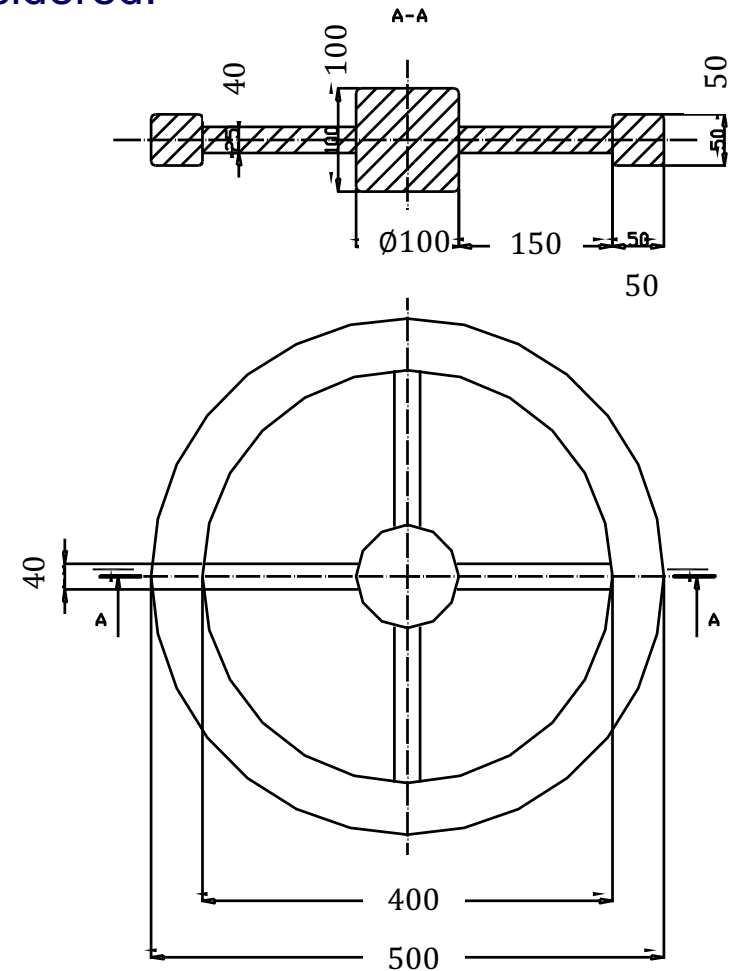
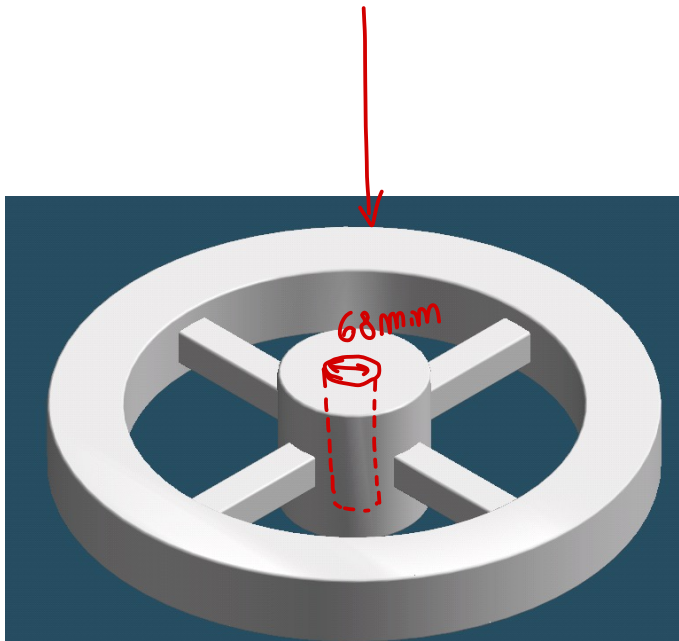
$\downarrow$
from M_R we design the RISERS



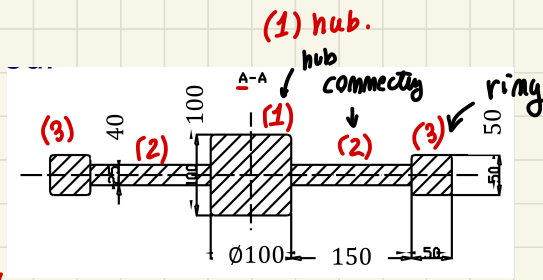
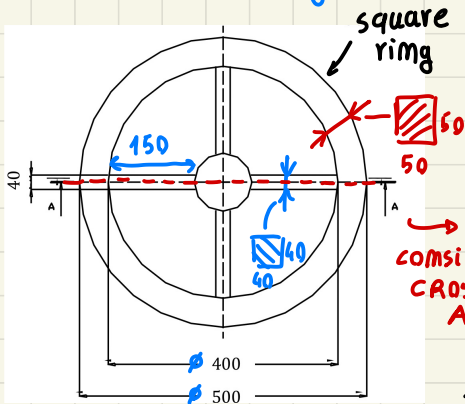
Example 4

Study the **directional solidification** of the casting shown in the figure.

Knowing that a 68 mm diameter hole will be drilled in the hub, if needed an up to 64 mm diameter cylindrical core may be considered.



Technical drawings

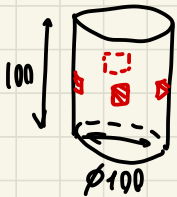


considering
CROSS-SECTION
A-A

DIRECTIONAL SOLIDIFICATION

to decompose the part ... look the section A-A

Body 1



$$V_1 = \frac{\pi}{4} 100^2 \cdot 100 = 785398 \text{ mm}^3$$

$$A_1 = 2 \frac{\pi}{4} 100^2 + \pi \cdot 100 \cdot 100 - 4 \cdot (40^2) = 40724 \text{ mm}^2$$

only of contact with
the HUB!

little
approx on
that
analysis



unreality, connection

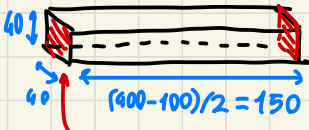
contact are curved
surface



$$H_1 = \frac{V_1}{A_1} = 19.29 \text{ mm}$$

Body 2

Just a simple squared parallelogram



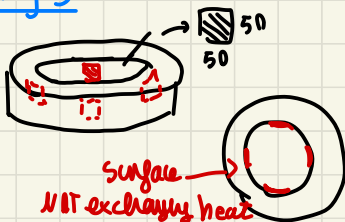
$$V_2 = 40^2 \cdot 150 = 240000 \text{ mm}^3$$

$$A_2 = 4 \cdot (40 \cdot 150) = 240000 \text{ mm}^2$$

$$\Rightarrow M_2 = \frac{V_2}{A_2} = 10 \text{ mm}$$

surface of connection,
NO HEAT exchange with HUB!

Body 3



$$V_3 = \frac{\pi}{4} 500^2 \cdot 50 - \frac{\pi}{4} 400^2 \cdot 50 = 3534292 \text{ mm}^3$$

$$A_3 = 2 \left(\frac{\pi}{4} 500^2 - \frac{\pi}{4} 400^2 \right) + \pi 500 \cdot 50 + \pi 400 \cdot 50 - 4 \cdot 50^2 =$$

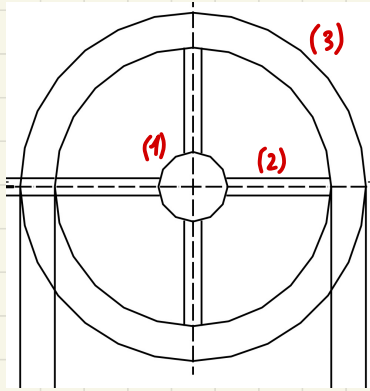
circle surface external cylinder int. cylinder

$$= 276343 \text{ mm}^2$$

$$\rightarrow M_3 = V_3 / A_3 = 12.79 \text{ mm}$$

$$\left\{ \begin{array}{l} M_1 = 19.29 \text{ mm} \\ M_2 = 10 \text{ mm} \\ M_3 = 12.79 \text{ mm} \end{array} \right\}$$

↓
same bodies
condition, just
look modulus



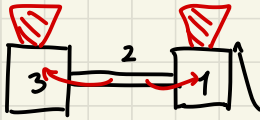
so

$$M_2 < M_3 < M_1$$

↓
FIRST to solidify
is B2.

↓
solidification
starts on the 4 Body 2
and continues along the
hub and the ring,
firstly ring solidify,
then the hub

theoretically



RISERS

solidification propagation

how we should position the risers...

TOP RISERS cause difficult side)

↓
BUT

comparing the M ratio (compare M ratio for solidification exchange flow)

$$\frac{M_3}{M_2} = \frac{12.8}{10} = 1.28 \in [1.1, 1.3] \text{ OK}$$

$$\frac{M_1}{M_2} = \frac{19.3}{10} = 1.93 \notin [1.1, 1.3] \text{ NO out of admissible range}$$

NO SHRINKAGE problem → B2 solidify and B1 stay melted
metal flow, 1 solidify -- OK

→ shrinkage @ SOLID STATE (cooling)

We have TOO HIGH cooling rate differences between B2 and B1

↓
creating internal stress on connecting points → ISSUE!

like fracture

↓
so looking on the EXERCISE text

HOLE in the hub

68mm HOLE @ the end

↑ doing a hole up to 64mm using a CORE → change casting shape

↑
issue during
cooling on SOLID
STATE

We know that in future with machining we will make a $\phi 68$ mm hole $\rightarrow$

we can do it right now with a CORE of 64 mm available

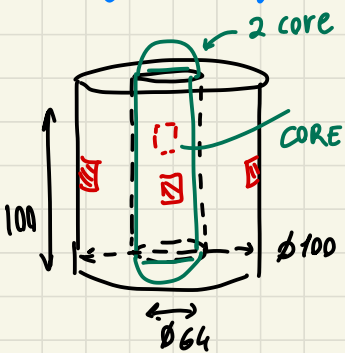
$\downarrow$
create hole smaller during CASTING (we will enlarge later)

making the CORE as $\phi 64$ mm

$\uparrow$ max CORE, so we minimize melt material and metal to machine

$\downarrow$
Body 1 changes

choose $\phi 64$ $\left\{ \begin{array}{l} \text{minimize melt metal} \\ \text{minimize metal to remove later} \end{array} \right.$



$$V_1^* = \left(\frac{\pi}{4} 100^2 - \frac{\pi}{4} 64^2 \right) 100 = 463699 \text{ mm}^3$$

(< 785398) reducing melt material to pour

$$A_1^* = 2 \left(\frac{\pi}{4} 100 - \frac{\pi}{4} 64^2 \right) + \pi 100 \cdot 100 + \pi \cdot 64 \cdot 100$$

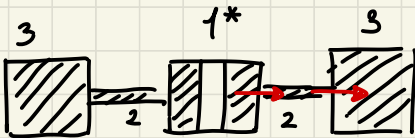
$$- 4 (40)^2 = 54396 \text{ mm}^2$$

(increasing the AREA)

$$H = \frac{V}{A} \downarrow \uparrow \rightarrow H \downarrow \uparrow$$

$$\rightarrow H_1^* = V_1^* / A_1^* = 8.52 \text{ mm}$$

Now... New behav.



$$8.52 < 10 < 12.8$$

$$H_1^* < H_2 < H_3$$

Cooking RATIO

$$\frac{H_3}{H_2} = 1.28 \text{ or as before}$$

Also directional solidification changes!

solidification starts on B1 and goes through B2 and then on B3

now $\downarrow$

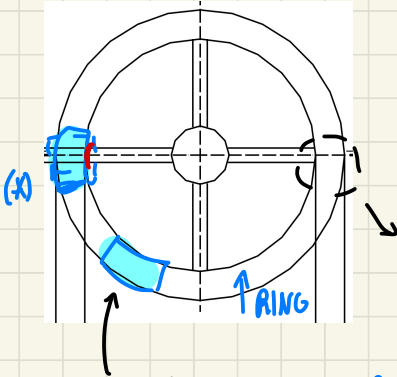
$$\frac{H_2}{H_1} = 1.174 \in [1.1, 1.3] \text{ OK!}$$

So now we have a CORRECT directional solidification

Bad Dimensions...
fracture @
solid
state

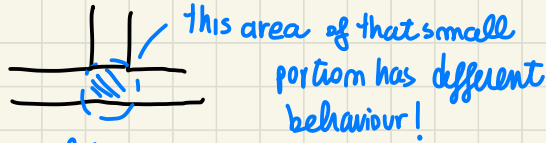
$$M_3 > M_2 > M_1$$

RISER to be positioned on Body 3
in contact



Where to position the RISERS on B3?

ATTENTION to connecting well T-geometries

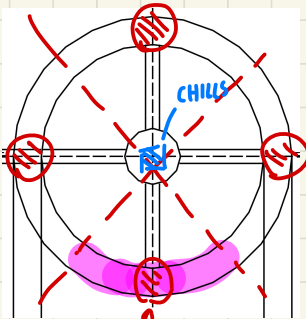


Same volume
but different
exchange area

em RING solidification is
NOT uniform → some points

→ to avoid possible issues...

I may position the risers on the NECK area



top
RISERS

(or I can use CHILLS)

putting a RISERS
just in connection
points →
single riser smaller
to feed 1/4 of total
cavity VOLUME

4 smaller Risers to
avoid any issues

@connecting POINT

here we risk to have a
problem cause changing behav

chills on part (1)
so it solidify
faster

↑
add chills
to increase
solidification
velocity

OR I can
rise by
using
different
Mold material
... different
exchange
condition

↓
more complex
mold to
shape different
MATERIAL...

It is possible
to play with
geometry and
tries to have

$$M_3 < M_2 < M_1 \rightarrow \text{so easy riser on B1.}$$

Design for CASTING → shape changing



Example 5



open top risers



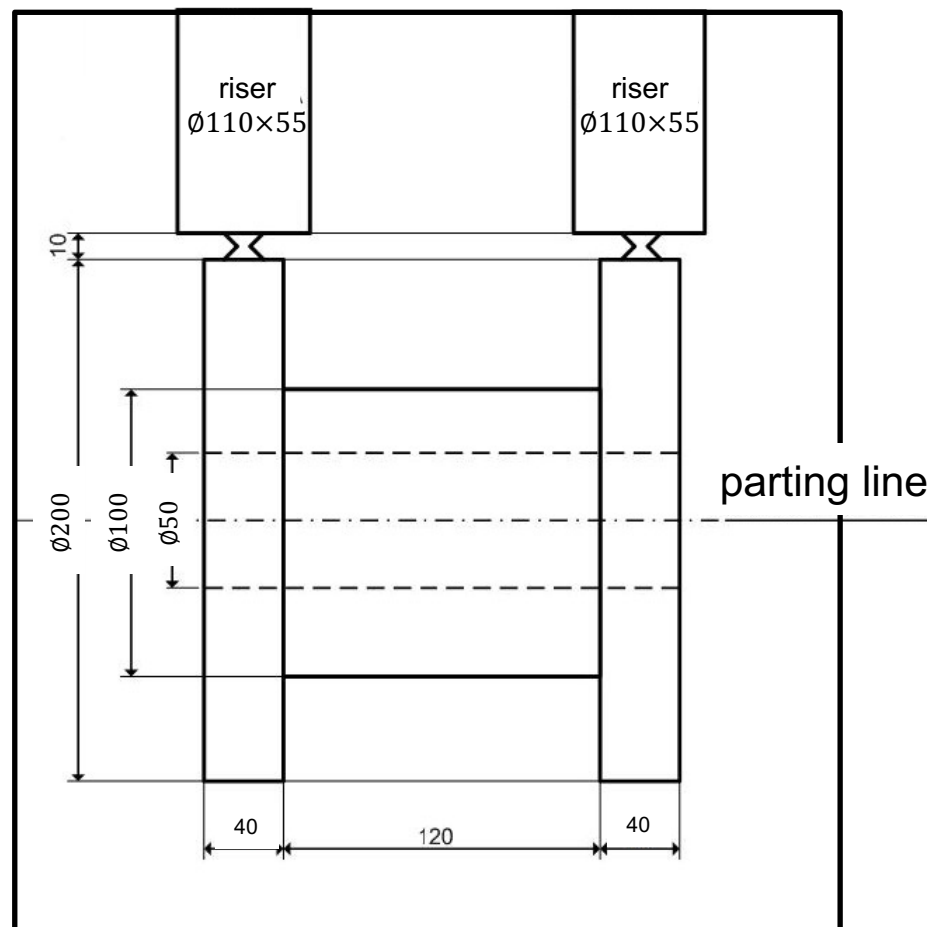
The figure shows a mould cavity with risers used in a sand casting process.

A middle gating system is considered with two gates, one runner and one sprue (middle gating system = gating system with gates on the parting line).

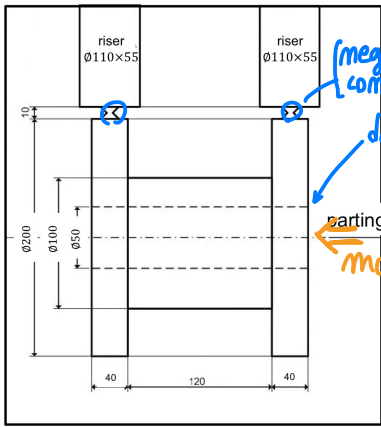
Design the gating system to ensure a mould filling time equal to 6 s.

Consider $A_s:A_r:A_g = 2:2:1$ and $c = 0,55$.

[friction loss
coefficient]



TOP cylindrical riser



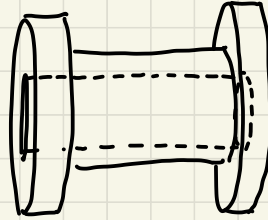
(negligible volume connecting elements)

dashed lines means a hole

parting line

material

The geometry to CAST is made of identical external cylinder and an internal one, all with a hole



GATING system design

MIDDLE GATING → material enters on parting line plane

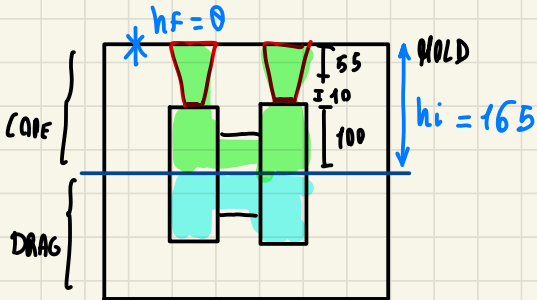
so for a middle gating system →

$$V = C \sqrt{2g H_{av}}$$

$h_i, h_f?$

$$H_{av} = \left(\frac{1}{\frac{V_1}{\sqrt{h_i}} + \frac{V_2}{\sqrt{h_i + h_f}}} \right)^2$$

V_1, V_2 are Volume ratio



h_i = distance between pouring cup (so top of core) respect middle

h_f = final distance where the liquid stay = 0

it is the highest point reached

as ending pouring wereach top riser so wereach reference

from pouring basin reference
TOP RISER → $h_f = 0$

IF pouring on parting line

portion of Volume in the DRAG / other portion on CORE

$V_1 \approx \text{DRAG}$ $V_2 \approx \text{COPE}$ respect the TOTAL

$$V_{WP} = 2 \frac{\pi}{4} 200^2 40 + \frac{\pi}{4} 100^2 120 - \frac{\pi}{4} 50^2 200 = 3063053 \text{ mm}^3$$

↑ WORK PIECE ext cylinder int cylinder hole

$$V_R = 2 \cdot \frac{\pi}{4} 110^2 \cdot 55 = 1045365 \text{ mm}^3$$

↑ RISERS

Neglecting connecting points volume $\Sigma 10$

$$\downarrow V_{TOT} = V_R + V_{WP} = 4108418 \text{ mm}^3$$

$$\begin{cases} r_1 = \frac{V_{WP}/2}{V_{TOT}} = 0.373 \\ r_2 = \frac{V_{WP}/2 + V_R}{V_{TOT}} = 0.627 \end{cases} \rightarrow 1-r_1$$

← DRAG PORTION

← COPE PORTION

so, once evaluated all that values...

$$H_{AV} = \left(\frac{1}{\frac{0.373}{\sqrt{155}} + \frac{0.627}{\frac{\sqrt{155} + \sqrt{0}}{2}}} \right)^2 = 62.33 \text{ mm}$$

use in [mm] to have same dimension

↓

$$V = 0.55 \sqrt{2 \cdot 9.81 \frac{62.33}{1000}} = 0.607 \frac{\text{m}}{\text{s}} < 1 \frac{\text{m}}{\text{s}}$$

↑ VELOCITY IN BOTTLENECK

OK fine (WE WANT NOT TOO FAST V)

known this...

Design GATING SYSTEM to have $T_{HF} \approx 10\text{s}$

evaluated A_{GMIN}

cause $A_s : A_R : A_G = 2 : 2 : 1$

from the ratio

$$A_s = A_R = 2 A_G = 2256 \text{ mm}^2$$

↑ BOTTLENECK is the smallest

area on the system! → GATE

MINIMUM AREA EQ GUARANTEES THE TARGET!

is the bottleneck

carefull with measure unit

$$A_{BN} = A_G$$

↓

$$\parallel Q = A_{BN} V \parallel$$

$$\textcircled{Q} \geq Q_{min} = \frac{V_{TOT}}{T_{HF}}$$

POURING RATE

greater than a minimum, target T_{HF}

$$A_{G,MIN} = \frac{V_{TOT}}{T_{HF} \cdot V} = \frac{4108418}{10 \cdot 0.607} = 1128 \text{ mm}^2$$

↑



Example 6



The pattern shown will be used to create a **cavity in sand casting**.

A **bottom gating system** has to be design to ensure a mould filling time equal to 5 s and $A_s:A_r:A_g = 3:3:2$. The height of the cope is equal to 75 mm.

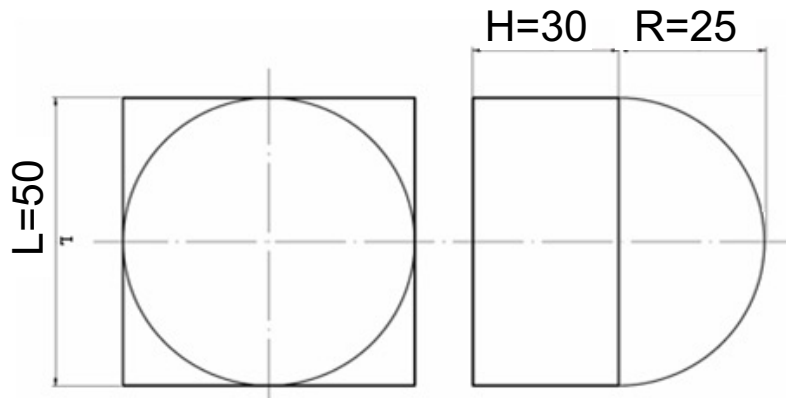
- a) **Design the gating system** considering $c = 0,5$ and a **blind riser on the parting line** with a volume equal to 50 cm^3 . *Riser NOT higher than cavity*
- b) If a gate has a cross-section equal to $0,25 \text{ cm}^2$, complete the **detailed design of the gating system** and determine the **actual mould filling time**.

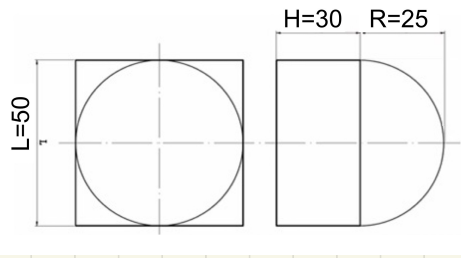
↑
Typical
exam question

a) DESIGN

b) given possible
design → evaluate

↑
INDEPENDENT
Questions (a), (b)

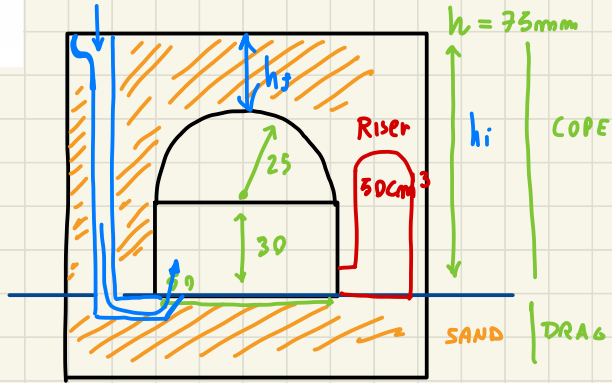




→ BOTTOM Gating system
= parting line in the Bottom
all on the COPE

blind side
RISER, usually
with high
smaller
than the cavity

↓
to design the
BOTTOM GATING SYSTEM
a)



$$A_s : A_R : A_G = 3 : 3 : 2$$

$$T_{HF} \leq 5s \text{ (GOAL)}$$

$$C = 0.5$$

b) given
 $a_G = 0.25 \text{ cm}^2$ (cross-section of ONE gate)

① as usual:

$$T_{HF} = \frac{V_{TOT}}{A_{BN} \cdot V}$$

Q → (A_{BN} · V)

total volume to fill

$$\text{Where } A_{BN} = \frac{V_{TOT}}{T_{HF} \cdot V}$$

(MIN) (MAX)
T_{HF} = 5s

to answer, we evaluate V:

$$V = C \sqrt{2g H_{AV}}$$

FRICION LOSSES → C
BOTTOM GATING SYS. → H_{AV}

$$H_{AV} = \left(\frac{\sqrt{h_i} + \sqrt{h_f}}{2} \right)^2$$

to evaluate h → $h_i = h_{COPE}$ ← because material

$h_i = 75 \text{ mm}$ flow from top to parting line, so
for the h_{COPE}, start from here to
reach cavity

While h_f depends on
final material height (cause riser usually smaller than cavity) → it is h_f ^{report pattern}

$$\begin{cases} h_f = 75 - (30+25) = 20 \text{ mm} & (\text{final height reached by material}) \\ h_i = 75 \text{ mm} & \text{respect top} \end{cases}$$

$$\Downarrow H_{AV} = \left(\frac{\sqrt{75} + \sqrt{20}}{2} \right)^2 = 43.115 \text{ mm}$$

$$V = C \sqrt{2g H_{AV}} = 0.5 \sqrt{2 \cdot 9.81 \frac{43.115}{1000} [\text{m}]} = 0.46 \frac{\text{m}}{\text{s}} \leq \frac{1 \text{ m}}{\text{s}}$$

$$V_{TOT} = V_{WP} + V_{RISER} = 157725 \text{ mm}^3$$

$\uparrow$
 WORK
PIECE
 (to fill of material)

experience based
rule, we may
stay $\leq 1 \text{ m/s}$ to
reduce erosion
issue $\rightarrow$ impurities

$$\begin{cases} V_{WP} = 50^2 \cdot 30 + \frac{1}{2} \frac{4}{3} \pi 25^3 = 107725 \text{ mm}^3 \\ V_{RISER} = 50 \text{ cm}^3 = 50000 \text{ mm}^3 \end{cases}$$

$$A_{BN} = \frac{157725}{5 \cdot 0.46 \times 1000} = 68.6 \text{ mm}^2 = A_G$$

the bottle neck
is the GATE

{

NOTICE that...

by ROUNDING.

$\downarrow$
 Considering $H_{AV} = 43.115 \approx 43 \text{ mm}$

$\downarrow$
 $V = 0.4593 \text{ m/s}$

$\uparrow$
 and on CASTING
 Problems $\rightarrow$ found $A_{BN} \rightarrow$ you have preshape channels to insert not
 shaped perfectly as wanted

does NOT
have this
much impact

here rounding HAS NO PROBLEM!

(b) if fixed $a_G = 0.25 \text{ cm}^2$ area of single GATE
requiring $A_G = 68.6 \text{ mm}^2$

$$N_G = \left\lceil \frac{A_G}{a_G} \right\rceil = \left\lceil \frac{68.6}{25} \right\rceil = \lceil 2.744 \rceil = 3 \text{ gates at least}$$

$\uparrow$ With $N_G = 2$ too
 small CROSS-SECT.
 I can't obtain
 THE
 required!

We will obtain a bigger AREA and a real HOLD FILLING TIME of

↓

$$T_{MF} = \frac{V_{TOT}}{A_{G \cdot U}} = \frac{157725}{\underbrace{3.25 \cdot 460}_{No. A_G}} = 4.57s < 5s \quad \underline{OK}$$

fill earlier, OK

←

bring more material per unit of time

choose a_G that we have preprocessed GATE

$$a_G = 25 \text{ mm}^2$$

↓

$$A_G = 75 \text{ mm}^2$$

$$A_S = A_R = \frac{3}{2} \cdot 75 = 112.5 \text{ mm}^2$$

↑
TOTAL

{ full GATING system design! }

we will have to choose a sprue with a value around that required A_S

Bottleneck

↑

is the constrain ...

so $A_G \geq 75 \text{ mm}^2$ or $T_{MF} \uparrow$

so also A_S, A_R would be ceil, AROUND that value!



Example 7

Vertical DRAG Cope...
You can do that keeping the
two elements close
when pouring



POLITECNICO
DI MILANO

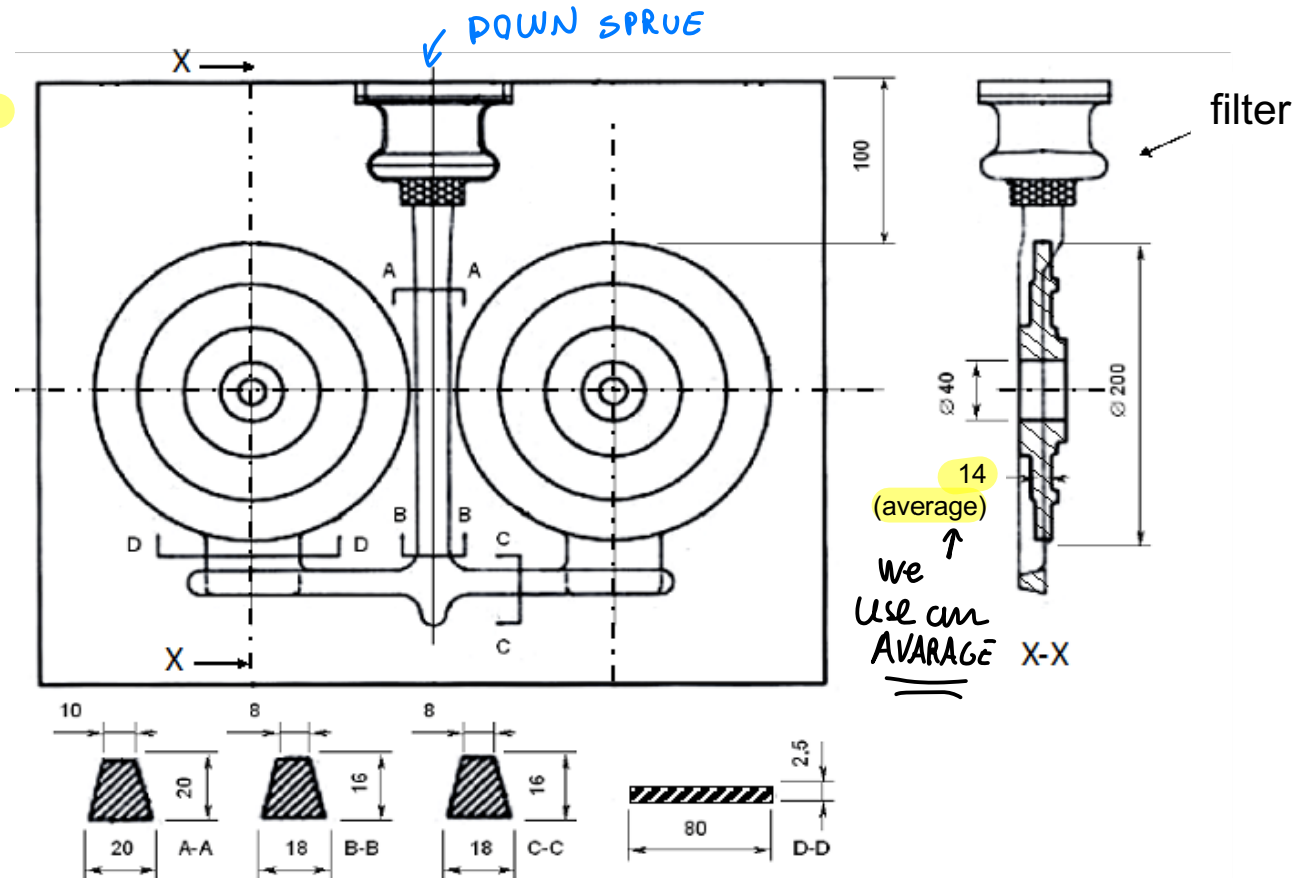
A foundry is casting brake discs for the automotive sector.

The figure shows the pattern used to shape a mould with vertical parting line.

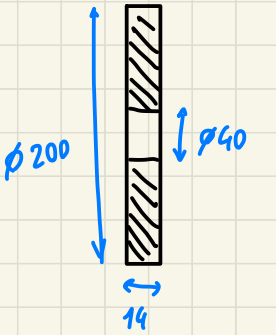
Consider an average thickness of the disc equal to 14 mm, two blind risers with a volume equal to $1,5 \cdot 10^5 \text{ mm}^3$ each and a height smaller than 100 mm.

Determine the V/A ratio for
the discs. MODULUS of DISK

Considering a disc volume
equal to $5 \cdot 10^5 \text{ mm}^3$,
compute the actual mould
filling time ($c = 40\%$) and
compare it with a time
constraint equal to 10 s.



considering the "AVERAGE"



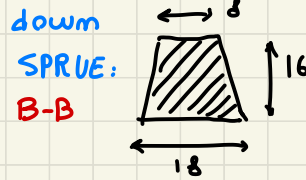
X-X CROSS SECTION

we prefer to consider these approx of real geometry

+ 2 BLIND RISERS

$$V_R = 2 \times 1.5 \cdot 10^5 \text{ mm}^3$$

$h_R < 100 \text{ mm}$
(Below cavity of the disk)

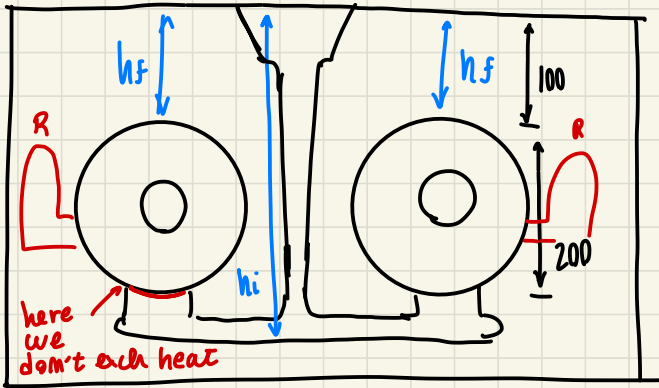
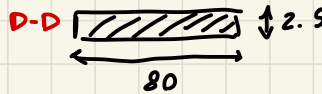


Rummers (2 Rummers)

because 2 cavity $\rightarrow$ 2 Rummers

C-C equal to sprue cross-section

Gates



BOTTOM GATING system
as I can see from the gate

$$h_i = 300 \text{ mm}$$

a) V/A

b) $V_{\text{disk}} = 5 \cdot 10^5 \text{ mm}^3$

(with that) $C = 0.4$
design $\rightarrow T_{MF} \leq 10s$

a) from our disk approx

exchange Area



$$V_{\text{disk}} = \frac{\pi}{4} 200^2 \cdot 14 - \frac{\pi}{4} 40^2 \cdot 14 = 422230 \text{ mm}^3$$

$$A_{\text{disk}} = 2 \left(\frac{\pi}{4} 200^2 - \frac{\pi}{4} 40^2 \right) + \pi 200 \cdot 14 + \pi 40 \cdot 14 = 70874 \text{ mm}^2$$

CIRCULAR RING

$$M_{\text{disk}} = V_{\text{disk}} / A_{\text{disk}} = 5.96 \text{ mm}$$

Not relevant difference

$A_{\text{disk}} = 70874 - 80 \times 2.5 = 70674 \text{ mm}^2$
 $\rightarrow M^* = 5.97 \text{ mm}$

NOTICE $\rightarrow$ here even Gate Area does NOT exchange heat

MINIMUM CHANGE!

Impact usually A_0 can be neglected and even the connection A with the RISER $\rightarrow$ negligible!

(b) BOTTOM GATING

$$v = C \sqrt{2g H_{AV}} \quad H_{AV} = \left(\frac{\sqrt{h_1} + \sqrt{h_2}}{2} \right)^2 = \left(\frac{\sqrt{300} + \sqrt{100}}{2} \right)^2 = 186.6 \text{ mm}$$

$$v = \underbrace{(0.4)}_{\substack{\text{LOT OF} \\ \text{LOSSES due to all} \\ \text{charging directions!}}} \sqrt{2 \cdot 9.81 \cdot \frac{186.6}{1000}} = 0.765 \frac{\text{m}}{\text{s}} \leq 1 \frac{\text{m}}{\text{s}} \text{ GOOD } v!$$

Which one is the Bottle neck? A_{BN}

from the full known Gating system design on channel p.o.v

$$\rightarrow A_s = (18+8) \cdot \frac{16}{2} = 208 \text{ mm}^2 \leftarrow \text{BOTTLE NECK } A_{BN}$$

TOTAL AREA $A_R = 2 \cdot A_s = 416 \text{ mm}^2$ (2 channels identical to the sprue)

$$A_0 = 2 \cdot a_0 = 2 \cdot 80 \cdot 2.5 = 400 \text{ mm}^2$$

$$\text{So... } T_{MF} = \frac{2 \cdot (500000 + 150000)}{208 \cdot 765 \left[\frac{\text{mm}}{\text{s}} \right]} = 8.17 \text{ s} < 10 \text{ s}$$

$V_{TOT} = V_{WP} + V_R$
OK less than CONSTRAINT



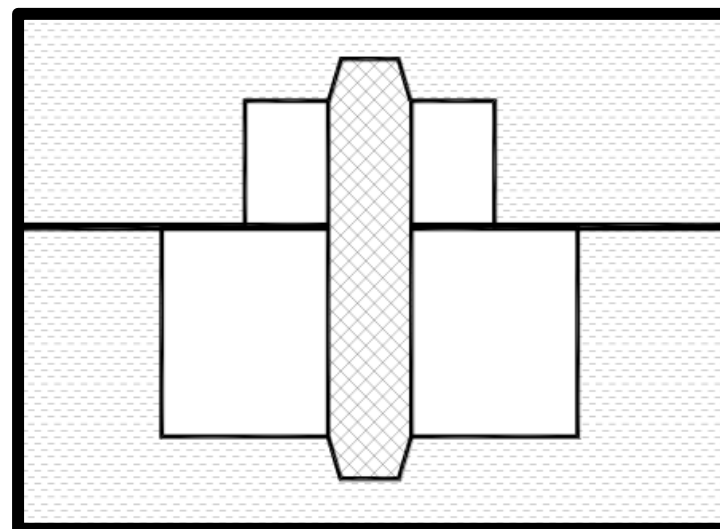
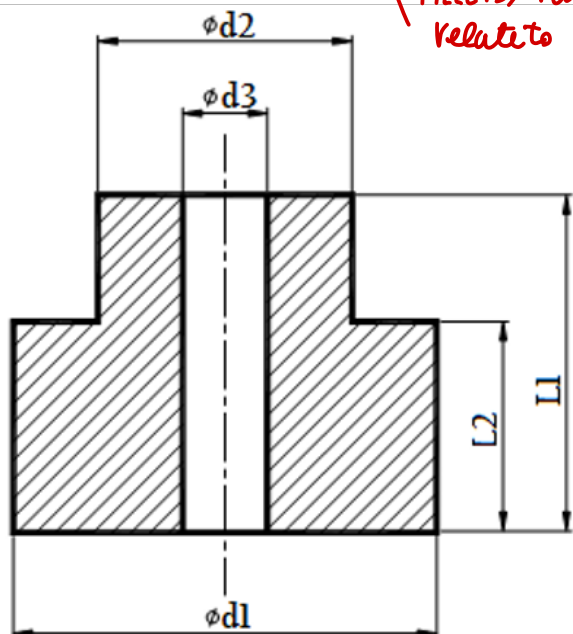
Example 8

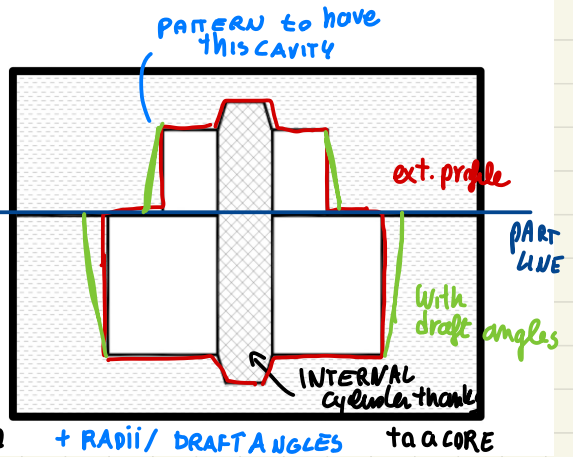
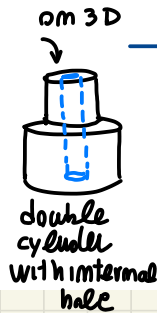
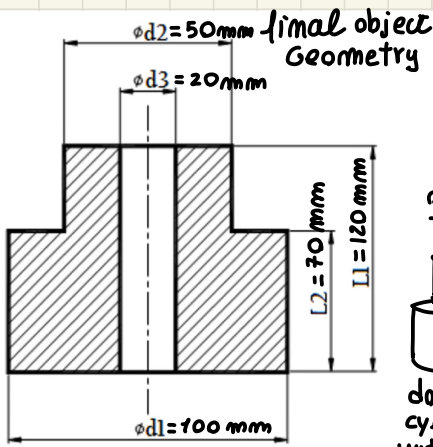


The figure shows a component to be obtained through a sand casting and machining processes. The component will be made of grey cast iron. The pattern will be made of wood. Dimensional data: $d1 = 100$ mm, $d2 = 50$ mm, $d3 = 20$ mm, $L1 = 120$ mm, $L2 = 70$ mm.

Design the pattern and the core respecting machining allowance (4 mm), draft angles (4°), fillets/radii (3 mm), and shrinkage allowance (1%). Suppose that all surfaces need to undergo subsequent machining.

FILLETS/radii depends on convexity, "Radii" relative to smooth





consider machining
+ shrinkage allowance

$$m.a = 4\text{ mm}$$

$$s.a = 1\%$$

DESIGN THE PATTERN

$$d_1^p = (100 + 2 \cdot 4) (1 + 0.01) = 108 \cdot 1.01 = 109.08 \text{ mm} \rightarrow 110 \text{ mm}$$

external diam

at least we remove later

ROUND TO integer with more material

external to add

$$d_2^p = (50 + 2 \cdot 4) 1.01 = 58 \cdot 1.01 = 58.58 \text{ mm} \rightarrow 59 \text{ mm}$$

$$d_3^p = (20 - 2 \cdot 4) 1.01 = 12 \cdot 1.01 = 12.12 \text{ mm} \rightarrow 12 \text{ mm}$$

pattern inside (internal diam)
for conical print

INTERNAL
room $\rightarrow$ we floor

$$L_1^p = (120 + 4 + 4) \cdot 1.01 = 128 \cdot 1.01 = 129.28 \text{ mm} \rightarrow 130 \text{ mm}$$

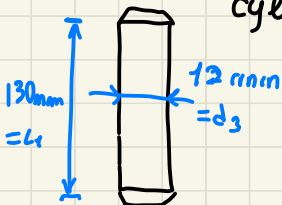
external
dimension

$$L_2^p = (70 + 4 + 4) \cdot 1.01 = 78 \cdot 1.01 = 78.78 \text{ mm} \rightarrow 79 \text{ mm}$$

PATTERN DIMENSION

CORE

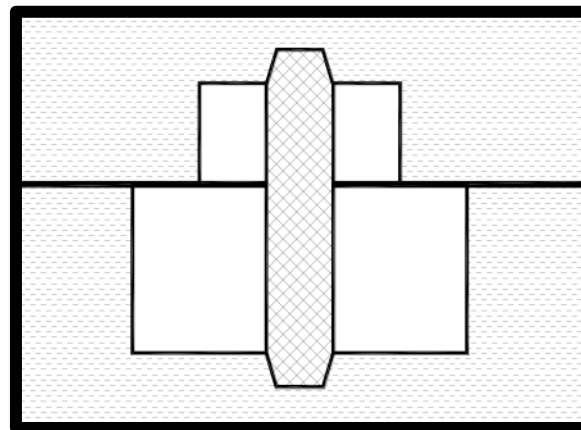
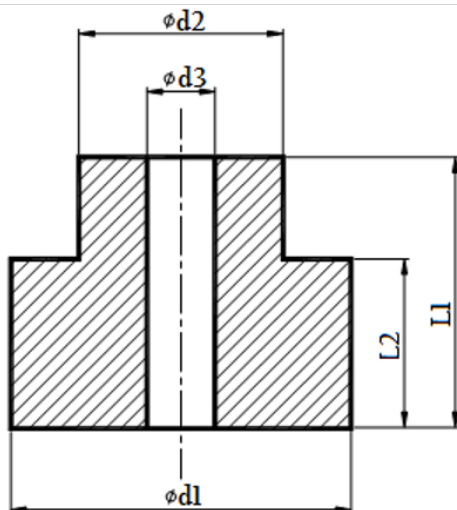
cylinder + core print



during solid shrinkage
the CORE is destroyed,
should NOT constrain the CORE!

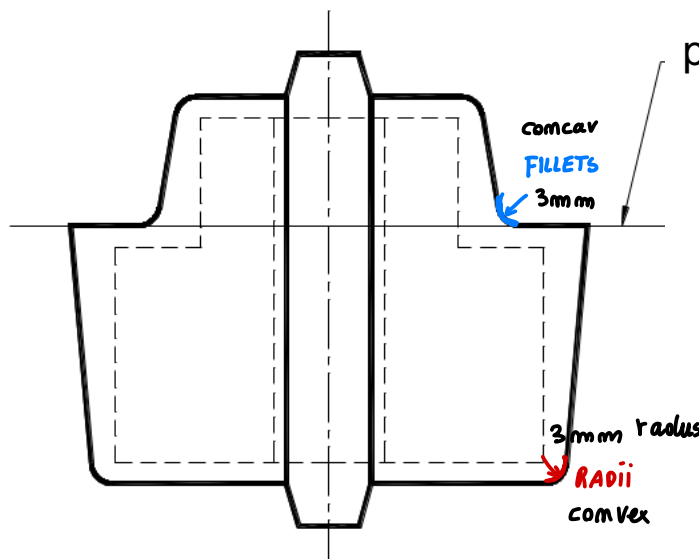


Example 8 - Solution



(Better)
that can be
removed later!
↑

To fully
answer, we
also need
to specify
how shape
change →



parting line

REAL
PATTERN

$$\left\{ \begin{array}{l} m.a \\ + \\ s.a \\ + \\ R_{add} \\ + \\ Draft \end{array} \right\}$$

NOT
to compute
it changes the
result!

(We neglect it,
but it should
be evaluated)

↑ adding material



Example 9

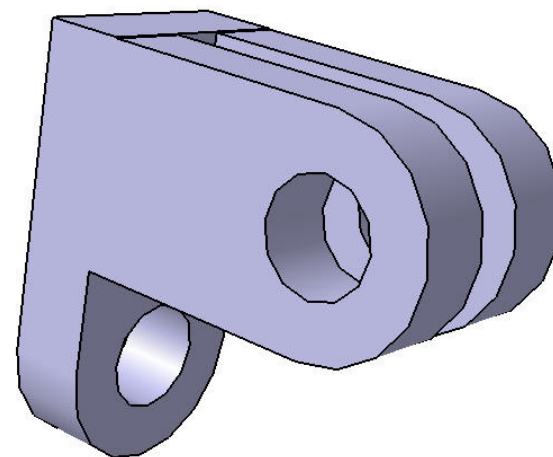
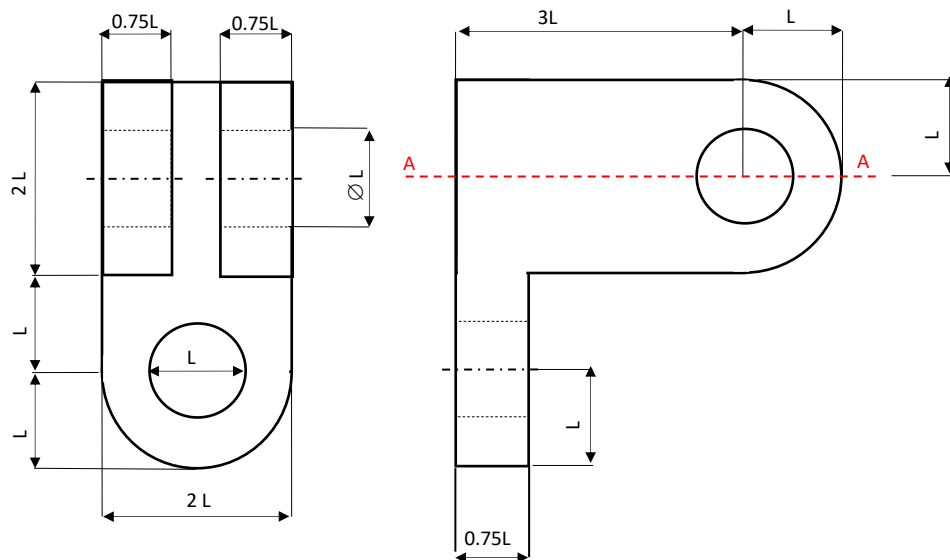


The figures show a component to be obtained through a sand casting. The plane A-A is chosen as the parting line of the mould.

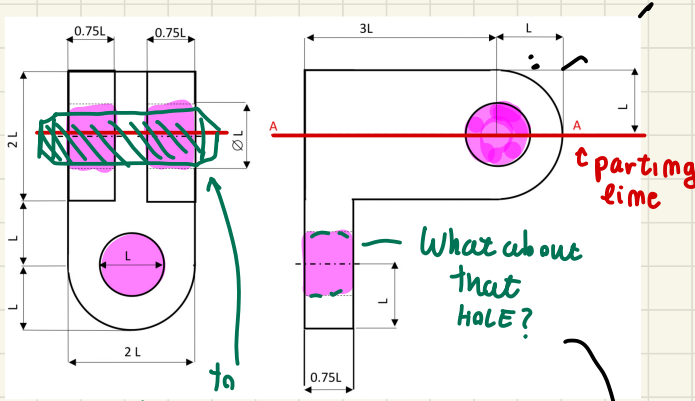
Design a middle gating system to ensure a mould filling time not greater than 10 s (do not consider risers and shrinkage allowance).

Data: $L = 2$ cm, $H_{\text{cope}} = 2L$, $H_{\text{drag}} = 3L$, $A_s:A_r:A_g = 4:2:1$, $c = 0,5$.

(BOTTLE NECK) *small c loss* *FINAL OBJECT*



thinking about the PARTING LINE. =>

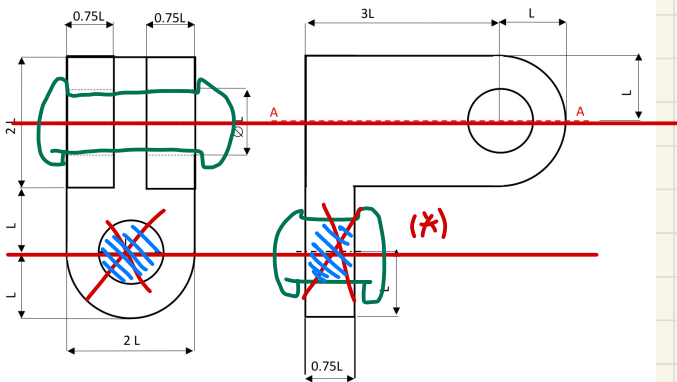


to obtain it, I can create a space to position the CORE for this holes

We should have a CORE PRINT TO position the CORE, or fill directly the cavity

BUT to remove the DRAG part of the PATTERN, the core print will destroy portion of the sand...

to do the second HOLE by a core, we should have a situation of 3 element mold → shaping bottom, middle / top and then put all together



3 part mold

↓
IT IS possible,
BUT HERE
WE WANT
JUST 2 section
DRAG / CORE

SO (*) that hole cannot be shaped during casting

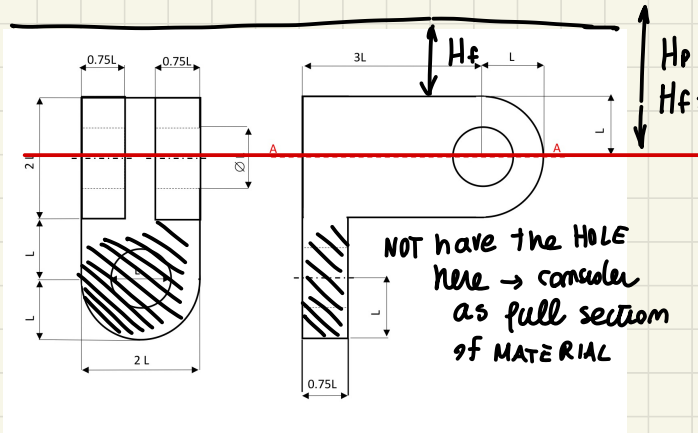
that hole will be done later by machining → postpone hole making → we consider that section FULL!

{ all holes can be done during CASTING ? }

by choosing that PARTING LINE...

↓
to have the internal holes with the chosen parting line

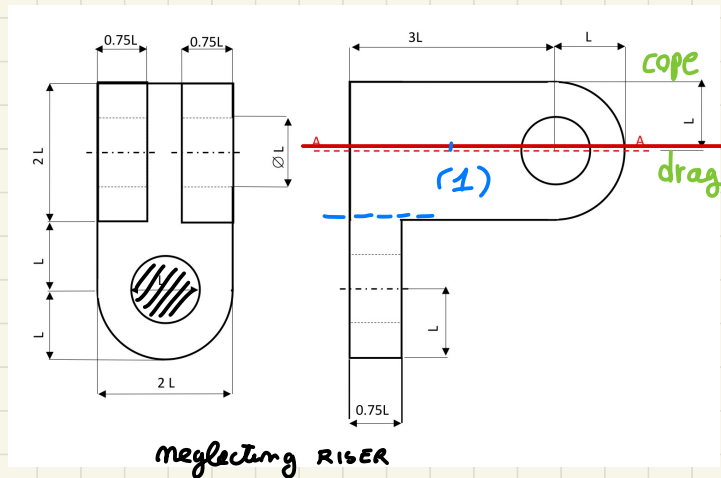
adding material as machine allowance → remove later


$$H_p = 2 \cdot L = 40 \text{ mm}$$

$H_f = 20 \text{ mm}$ $\rightarrow$ without
risers, but
risers in the side

describing all
the elements

DESIGN A MIDDLE GATING SYSTEM


$$L = 2 \text{ cm} = 20 \text{ mm}$$

$$V_1 = 15 \cdot 40 \cdot 40 + 2 \cdot 15 \cdot 40 \cdot 45 + 2 \left[\left(\frac{1}{2} \pi \cdot 20^2 \cdot 15 \right) - \left(\frac{\pi}{4} 20^2 \cdot 15 \right) \right] = 87425 \text{ mm}^3$$

$$V_2 = 20 \cdot 40 \cdot 15 + \frac{1}{2} \left[\frac{\pi}{4} 20^2 \cdot 15 \right] = 21425 \text{ mm}^3$$

$$V_{COPE} = V_1 / 2 = 43712.5 \text{ mm}^3$$

$$V_{\text{DRAG}} = V_1/2 + V_2 = 65137.5 \text{ mm}^3$$

$$\begin{aligned} \overline{V}_{TOT} &= \overline{V}_1 + \overline{V}_2 = \\ &= 1080850 \text{ mm}^3 \end{aligned}$$

$$r_{\text{core}} = \bar{V}_{\text{core}} / V_{\text{TOT}} = 0.4016$$

$$v_{drag} = V_{DRAG} / V_{TOT} = 0.5984 \quad \rightarrow \quad v = C \sqrt{2g H_{AV}} = 0.5 \sqrt{2 \cdot 9.81 \cdot \frac{35}{1000}} =$$

$$H_{Av} = \left(\frac{1}{\frac{0.5984}{\sqrt{40}} + \frac{0.4016}{\frac{\sqrt{40} + \sqrt{20}}{2}}} \right)^2 = 35 \text{ mm} \quad \uparrow = 0.414 \frac{\text{m}}{\text{s}} \leq 1 \frac{\text{m}}{\text{s}}$$

$$A_{BU} = A_G = \frac{108850}{10s \cdot 0,414 \cdot 1000} = 26.29 \text{ mm}^2$$

$$A_{\uparrow} = 2 A_G = 52.58 \text{ mm}^2$$

Rummers

min possible A

$$A_s = 4 A_G = 105.16 \text{ mm}^2$$

SPAU